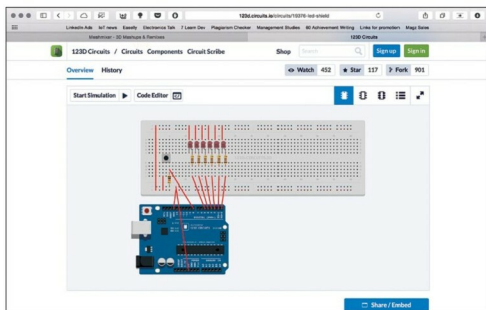




123D Circuits by Autodesk

that and optimise it. The benefit of this is that, in the bigger perspective of the system you do not need every component to be ready before you test the whole thing. You can actually have multiple components and then just one component can be refined.

EDA has been well-adopted by chip designers to design and analyse entire semiconductor chips over the last few decades. Jason Guo, senior technical staff member, Fairchild Semiconductor, says, "However, system-level design tools, such as tools for a switched mode power supply design, are not readily available, mainly due to system-level complexities. Recently, some advanced Web based electronic design/simulation tools such as Fairchild's Power Supply WebDesigner (PSW) have been introduced to enable system-level design of switched mode power supplies."

These tools are typically driven by knowledge based automation for every decision making point of a design flow, so that even a junior designer can complete a senior-level design in minutes. Web based simulation is used to bridge conventional mismatching gaps between a paper design and hardware performance to further secure a working design.

Announced in February 2015, Altera and Mentor Graphics have teamed up together to provide virtual platforms that contain simulation models of ARM processor subsystems featured in Altera's SoC field programmable gate array (FPGA) families. This simplifies embedded software development and reduces the need for expensive hardware by enabling large development teams to perform application development and software regression testing in a virtual environment based on the chip.

Unconventional labs and education

The three kinds of laboratories in Indian engineering institutions are conventional labs, virtual labs and remote labs. In most engineering institutions the laboratory infrastructure is significantly underutilised due to factors such as security, timings and available working days.

Stated in the paper 'A Distinctive Approach to Enhance the Utility of Laboratories in Indian Academia' presented at the 2015 International Conference on Remote Engineering and Virtual Instrumentation, authors go on to explain how remote labs can be used to bring awareness about engineering technologies and software

in rural schools and colleges in India.

Kalyan Ram B., co-author of the above paper and co-founder and CEO of Electrono Solutions Pvt Ltd, spoke at EFY's Electronics Rocks 2014 conference on an innovative product named Remote Labs. Essentially a laboratory controlled over the Internet, it is designed exclusively for learning and monitoring any experiment integrated into the college lab. It can also be used by industries for monitoring electronics integrated with the lab.

Another interesting tool is SPARKS (simulations for performance assessments that report on knowledge and skills) software—a virtual lab that targets the beginner side of engineering. It teaches introductory electronics via a virtual breadboard, components and test equipment. The tool monitors, records and interprets student performance. If a student makes a mistake, the computer will point it out; if a critical step was left out, the software will observe and report that too.

Another tool with similar functionality is Sakshat Virtual Labs started by Ministry of Human Resource and Development. The team states at their website that the objective of this lab is to perform virtual experiments in basic electronics and yet have a close-to-real-life experience. The tool is designed in such a way that students get a feel of a real lab, as its equipment, components and behaviour are mimicked by the system.

Virtual Labs and SOLVE

Virtual Labs initiative was set up with the primary objective of experiential learning of the basic concepts in engineering technology and science. It also aims to make the labs available at the learner's time, pace and place with the aim to make it easier to conduct experiments. The virtual lab set up at NITK, Surathkal,